

SLIDE 1 — What are Basic Array Operations?

Definition:

Array operations are actions performed on array elements, such as **sum**, **average**, **maximum**, and **minimum**.

Simple idea

Array → **Process** → **Result**

Example:

```
int numbers[5] = {10, 20, 30, 40, 50};
```

We can calculate:

- **Sum** → Add all values
- **Average** → Find the mean
- **Maximum** → Find largest value
- **Minimum** → Find smallest value

 **Remember: Array + Loop → Easy calculations!**

SLIDE 2 — Finding the Sum

Example

```
int numbers[5] = {10, 20, 30, 40, 50};  
int sum = 0;
```

```
for (int i = 0; i < 5; i++) {  
    sum = sum + numbers[i];  
}
```

Explanation:

10 + 20 + 30 + 40 + 50

↓
150

Output:

Sum = 150

★ **Remember**

sum = sum + array[i]

SLIDE 3 — Finding Average

Average = Sum ÷ Number of Elements

Example

```
int numbers[5] = {10, 20, 30, 40, 50};  
int sum = 0;
```

```
for (int i = 0; i < 5; i++) {  
    sum += numbers[i];  
}
```

```
float average = (float)sum / 5;
```

Explanation:

Sum = 150

Elements = 5

Average = 150 / 5
= 30

Output:

Average = 30

🧠 **Remember: Average = Total ÷ Count**

SLIDE 4 — Maximum & Minimum

Maximum

Finds the **largest value**.

```
int max = numbers[0];

for (int i = 1; i < 5; i++) {
    if (numbers[i] > max)
        max = numbers[i];
}
```

Maximum = 50

Minimum

Finds the **smallest value**.

```
int min = numbers[0];

for (int i = 1; i < 5; i++) {
    if (numbers[i] < min)
        min = numbers[i];
}
```

Minimum = 10

Basic Pattern

START → COMPARE → UPDATE → REPEAT

SLIDE 5 — RECAP

Basic Array Math!

Flow

ARRAY



LOOP
↓
PROCESS VALUES
↓
CALCULATE
↓
RESULT

5 Things to Remember

1. Sum adds all elements
2. Average = Sum ÷ Number of elements
3. Maximum finds the largest value
4. Minimum finds the smallest value
5. Loops help process every element

Final Example

```
int numbers[4] = {5, 10, 15, 20};  
Sum    → 50  
Average → 12.5  
Maximum → 20  
Minimum → 5
```

Memory Trick

SUM → ADD 

AVERAGE → DIVIDE 

MAX → BIGGEST 

MIN → SMALLEST 

QUESTIONS

1. Find the Sum

Complete the code to calculate the sum of all elements.

```
int numbers[5] = {10, 20, 30, 40, 50};
```

```
int sum = 0;
```

```
for (int i = 0; i < 5; i++) {
```

```
    sum = sum + _____;
```

```
}
```

```
printf("%d", sum);
```

💡 Hint: Add the current array element to **sum**.

Answer: **numbers[i]**

2. Find the Average

Complete the code to calculate the average of the array elements.

```
int numbers[5] = {10, 20, 30, 40, 50};
```

```
int sum = 0;
```

```
for (int i = 0; i < 5; i++) {
```

```
    sum += numbers[i];
```

```
}
```

```
float average = (float)sum / ____;
```

```
printf("%.1f", average);
```

💡 Hint: Average = Sum ÷ Number of elements.

Answer: 5

3. Find the Maximum

Complete the code to find the largest value in the array.

```
int numbers[5] = {10, 50, 30, 20, 40};
```

```
int max = numbers[0];
```

```
for (int i = 1; i < 5; i++) {
```

```
    if (_____ > max)
```

```
        max = number[i] ;
```

```
}
```

```
printf("%d", max);
```

💡 Hint: When a larger value is found, update **max** with that value.

Answer: **numbers[i]**

4. Find the Minimum

Complete the code to find the smallest value in the array.

```
int numbers[5] = {25, 10, 40, 15, 30};
```

```
int min = numbers[0];
```

```
for (int i = 1; i < 5; i++) {  
    if (_____  
        min =numbers[i];  
}
```

```
printf("%d", min);
```

💡 Hint: When a smaller value is found, update min.

Answer: numbers[i]<min

5. Arrange the steps in the correct order:

- A. Get the result
- B. Access array element
- C. Start the loop
- D. Perform calculation

💡 Hint: Start → Access → Calculate → Result.

📖 Glossary: Calculation → A mathematical operation performed on values.

✅ Answer: C → B → D → A